Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva) (along with other discussion).
- Maths Lunch Hours, Mondays and Wednesdays, 1.00pm - 2.00pm. Bragg GR.18.
- My student hour: Tuesdays 1.00pm - 2.00pm, Bragg building foyer.

Exam preparation

Let n be a positive integer.

Let $R \subseteq \mathbb{Z} \times \mathbb{Z}$ be the relation defined on the integers so that $(a, b) \in R$ if and only if n divides $a - b$.

Prove that R is reflexive.

Revision

- If $b > 1$ is an integer, then every non-negative integer has a unique base-b representation.
- A base-b representation of m uses $\lfloor \log_b m \rfloor + 1$ digits.
- When we consider algorithms that process integers, we need the running time to be a function like (number of input digits)^c, where c is a fixed number. Such an algorithm is **polynomial-time**.
- If an algorithm is polynomial-time with respect to one base, then it is polynomial-time with respect to any base, so we often do not specify which particular base we are using.
- The usual algorithms for addition and multiplication are polynomial-time.
- How do we perform exponentiation in polynomial-time?

Quotients and remainders

Let m and n be positive integers. If q and r are integers such that $m = qn + r$ and $0 \leq r < n$, then we say q is the **quotient** and r is the **remainder** when we divide m by n.

Example. $52 = 6 \times 8 + 4$, so the quotient is 6 and the remainder is 4 when dividing 52 by 8.

Quotients and remainders

Theorem 9.9. Let m and n be positive integers. There are unique integers q and r such that $m = qn + r$, and $0 \leq r < n$.

Proof. We must prove q and r exist, and that they are unique. We leave the second part as an exercise. Assume for a contradiction that $m, n \in \mathbb{Z}^+$ do not have a quotient/remainder. We assume that amongst such pairs, m is as small as possible.

If $m < n$, then $m = 0 \times n + m$ where $0 \leq m < n$ so the quotient/remainder do exist.

So $m \not< n$, implying $m \geq n$.

Quotients and remainders

If $m = n$, then $m = 1 \times n + 0$ where $0 \leq 0 < n$ so the quotient/remainder exist, a contradiction.

There $m \neq n$, so $m > n$, implying $m - n$ is positive, and smaller than m.

The assumption on m, means when we divide $m - n$ by n, the quotient/remainder exist.

$m - n = q'n + r'$ where $0 \leq r' < n$.

Implying $m = (q' + 1)n + r'$ where $0 \leq r' < n$.

So the quotient/remainder does exist, a contradiction. We set $q = q' + 1$ and $r = r'$.

□

$m > n \qquad m = 1 \times n + (m - n)$

Quotients and remainders

Question. How do we find the quotient and remainder if we divide 2710 by 24?

We could repeatedly subtract 24 from 2710 and count the number of loops it takes to reduce to a number that is less than 24.

This does not lead to a polynomial-time algorithm!

2710 − 2400 = 310 ; $q \leftarrow 100$

310 − 240 = 70 ; $q \leftarrow 100 + 10 = 110$

70 − 24 = 46 ; $q \leftarrow 110 + 1 = 111$

46 − 24 = 22 ; $q \leftarrow 111 + 1 = 112$

Halt. $q = 112$, $r = 22$

Calculating quotient and remainder

Input: Base-b representations of $m = (m_s m_{s-1} \cdots m_1 m_0)_b$ and $n = (n_t n_{t-1} \cdots n_1 n_0)_b$ $(n \leq m)$.

Output: q and r such that $m = qn + r$ and $0 \leq r < n$.

$r \leftarrow m$
$i \leftarrow s - t$
$q \leftarrow 0$
while $r \geq n$ **do**
  while $r \geq nb^i$ **do**
    $r \leftarrow r - nb^i$
    $q \leftarrow q + b^i$
  end while
  if $i > 0$ **then**
    $i \leftarrow i - 1$
  end if
end while

"$\leftarrow$" means "is assigned"

$m = 110$ $n = 24$ $b = 10$

$r \leftarrow 110$ $i \leftarrow 3 - 2 = 1$

Is $r \geq 24 \times 10^1$

No.

Is $i > 0$. Yes.

$i \leftarrow i - 1 = 0$

Is $110 = r \geq nb^i = 24 \times 10^0$

Yes. $r \leftarrow 110 - 24 = 86$

$q \leftarrow 0 + 1 = 1$

Modular equivalence

Proposition 9.10. Let n be a positive integer. Define a relation, R, on the set of integers as follows: if a and b are integers, then (a, b) is in R if and only if n divides $a - b$. Then R is an equivalence relation.

Proof. Reflexive: We must prove that for an arbitrary a, $(a, a) \in R$. Since any integer divides 0, we have $n \mid (a-a)$, so $(a,a) \in R$.

Symmetry: We assume $(a,b) \in R$. This means $n \mid (a-b)$. Therefore $a-b = kn$ for some $k \in \mathbb{Z}$. We multiply both sides by -1 to obtain $b-a = (-k)n$. So $n \mid (b-a)$ and hence $(b,a) \in R$.

Modular equivalence

Transitive: Left as exercise.

(Hint. Use the fact that if n divides two numbers, it divides their sum).

Modular equivalence

We write $a \equiv b \pmod{n}$ to mean that n divides $a - b$.

We say that a and b are **equivalent modulo n**.

$2 \mid (3-1)$ so $3 \equiv 1 \pmod{2}$

$5 \mid (2-7)$ so $2 \equiv 7 \pmod{5}$

$12 \mid (5-5)$ so $5 \equiv 5 \pmod{12}$

Residues

Say that m and n are positive integers.

If $m = qn + r$, where $0 \leq r < n$, then r is called the **residue of m modulo n**.

We write m mod n for the residue of m modulo n.

Examples.

$12 \bmod 5 = 2$

$31 \bmod 7 = 3$

$3 \bmod 10 = 3$

$12 \bmod 6 = 0$

$13 \bmod 13 = 0$

Residues

A residue modulo n is always a number in the range $0, 1, 2, \ldots n-1$, so a residue is a potential remainder.

Producing m mod n from m is called **reducing** modulo n.

In programming languages, m mod n is often written `m % n`.

Note the difference between the notations

$$m \bmod n \quad \text{and} \quad a \equiv b \pmod{n}.$$

This represents a number: the remainder after dividing m by n.

This expresses a relation: it say n divides a-b.

Repeated squaring modulo n

In cryptography (sending secret messages), we often need to find $x^y \bmod n$, where y is a very large number.

Question. How do we calculate $x^{(y_n y_{n-1} \cdots y_2 y_1 y_0)_2} \bmod n$?

$$
\begin{aligned}
& x^{(y_n y_{n-1} \cdots y_2 y_1 y_0)_2} \bmod n \\
&= x^{2^n y_n + 2^{n-1} y_{n-1} + \cdots + 4y_2 + 2y_1 + y_0} \bmod n \\
&= x^{2^n y_n} x^{2^{n-1} y_{n-1}} \cdots x^{4y_2} x^{2y_1} x^{y_0} \bmod n \\
&= (x^{2^n})^{y_n} \cdots (x^4)^{y_2} (x^2)^{y_1} (x)^{y_0} \bmod n \\
&= ((x^{2^n})^{y_n} \bmod n) \cdots ((x^2)^{y_1} \bmod n)((x)^{y_0} \bmod n) \\
&= (x^{2^n} \bmod n)^{y_n} \cdots (x^2 \bmod n)^{y_1} (x \bmod n)^{y_0}
\end{aligned}
$$

mod n

Repeated squaring modulo n

We produce the sequence $(x \bmod n)$, $(x^2 \bmod n)$, $(x^4 \bmod n)$,... by repeatedly squaring and then reducing modulo n.

We keep a running product z and update z by multiplying with $x^{2^i} \bmod n$ (and then reducing modulo n) if $y_i = 1$.

Input: Base-2 representations of x and $y = (y_t y_{t-1} \cdots y_1 y_0)_2$.
Output: A representation of $z = x^y \bmod n$.

$z \leftarrow 1$
$p \leftarrow x \bmod n$
for $i = 0$ to t **do**
    if $y_i = 1$ **then**
        $z \leftarrow z \cdot p \bmod n$
    end if
    $p \leftarrow p^2 \bmod n$
end for

Repeated squaring modulo n

Question. How do we calculate 5^{45} mod 7?

Express 45 as $32+8+4+1 = (101101)_2$.

i	y_i	z	p
		1	$5 \bmod 7$
0	1	$1 \times 5 \bmod 7 = 5$	$5 \times 5 \bmod 7 = 4$
1	0	$5 \bmod 7$	$4 \times 4 \bmod 7 = 2$
2	1	$5 \times 2 \bmod 7 = 3$	$2 \times 2 \bmod 7 = 4$
3	1	$3 \times 4 \bmod 7 = 5$	$4 \times 4 \bmod 7 = 2$
4	0	$5 \bmod 7$	$2 \times 2 \bmod 7 = 4$
5	1	$5 \times 4 \bmod 7 = 6$	

Greatest common divisors

Remember that the notation $p|q$ means that p divides q.

Definition 9.13. Let m and n be positive integers. If d is an integer and $d|m$ and $d|n$ are both true, then d is a **common divisor** of m and n.

The largest common divisor is called the **greatest common divisor** of m and n. We denote it by $\gcd(m, n)$.

Example. The common divisors of 40 and 12 are:

$4, 2, 1, -1, -2, -4$, so

$\gcd(40, 12) = 4$.

How do we find $\gcd(m, n)$?

Proposition 9.14. Let m and n be positive integers where $m \geq n$. If $r = m \bmod n$, then $\gcd(m, n) = \gcd(n, r)$.

What is $\gcd(57, 40)$?

$57 \bmod 40 = 17$ so $\gcd(57,40)$

$= \gcd(40, 17)$

$40 \bmod 17 = 6$ $= \gcd(17, 6)$

$17 \bmod 6 = 5$ $= \gcd(6, 5)$

$6 \bmod 5 = 1$ $= \gcd(5, 1)$

$5 \bmod 1 = 0$ $= \gcd(1, 0) = 1$